

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

**COURSE NAME: Cloud Computing and
Big Data Analytics**

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO5: Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.

(BL5)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Articulation (Level 4)

Question Count: 10

Total Marks: 25

Code of Conduct

I A. Reddy Prasad (192472045) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Reddy Prasad

1. Explain the difference between HDFS and traditional file systems.

Answer:

HDFS (Hadoop Distributed File System) is designed to store very large amounts of data across multiple machines, whereas traditional file systems usually store data on a single computer or storage device.

Key differences:

- HDFS distributes files across multiple **DataNodes**.
- HDFS uses **replication** for fault tolerance.
- It is designed for **large files and sequential access**.
- Traditional file systems generally depend on a single server or storage system.
- HDFS can scale horizontally by adding more DataNodes.

Architecture:

Client → NameNode → Multiple DataNodes

2. Why is replication important in distributed storage systems?**Answer:**

Replication means keeping multiple copies of the same data on different nodes. It is important because individual machines can fail.

For example, with a **replication factor of 3**, three copies of a block are stored on different DataNodes.

Benefits:

- Provides **fault tolerance**.
- Improves **data availability**.
- Prevents data loss when a node fails.
- Can improve read performance because data can be read from different replicas.

Architecture:

Data Block → Node 1 + Node 2 + Node 3

3. How does MapReduce divide work between map and reduce phases?**Answer:**

MapReduce divides a large data-processing task into two main phases.

Map phase:

- Input data is divided into smaller blocks.

- Multiple Mapper tasks process these blocks in parallel.
- Each Mapper generates key-value pairs.

Example:

Input logs → Mapper → (Error, 1)

Reduce phase:

- The Shuffle and Sort stage groups similar keys.
- Reducers process the grouped values.

Example:

(Error, [1,1,1,1]) → Reducer → (Error, 4)

Architecture:

Input → Map → Shuffle & Sort → Reduce → Output

This parallel design allows Hadoop to process very large datasets efficiently.

4. What role does YARN play in large-scale Hadoop processing?

Answer:

YARN stands for **Yet Another Resource Negotiator**. It manages cluster resources and schedules applications in a Hadoop cluster.

Its main responsibilities are:

- Allocating **CPU and memory**.
- Scheduling applications.
- Managing application execution.
- Monitoring running applications.
- Handling failures and restarting tasks when necessary.

Main components:

- **ResourceManager:** Manages overall cluster resources.
- **NodeManager:** Manages resources on individual nodes.
- **ApplicationMaster:** Manages a particular application.

Architecture:

Client → ResourceManager → NodeManagers → Application Containers

5. Differentiate NAS and SAN in an enterprise data environment.

Answer:

NAS	SAN
Network Attached Storage	Storage Area Network
Provides file-level storage	Provides block-level storage
Accessed using file protocols	Accessed using storage protocols
Easier to manage	More complex
Common for shared files	Common for databases and enterprise applications
Usually uses Ethernet networks	Often uses dedicated high-speed storage networks

Simple example:

NAS is like a **shared network folder**, while SAN provides storage that appears to servers more like **local disks**.

6. What is the purpose of ETL in a big data pipeline?

Answer:

ETL means **Extract, Transform, Load**.

- **Extract:** Collect data from different sources.
- **Transform:** Clean, validate, filter, and convert the data.
- **Load:** Store the processed data in a target system such as HDFS or a data warehouse.

Example:

Database + CSV + API → Extract → Clean/Transform → HDFS/Data Warehouse

ETL ensures that data is **clean, consistent, and ready for analysis**.

7. How does Apache NiFi help in data integration workflows?

Answer:

Apache NiFi is a data-flow and integration platform used to move and process data between different systems.

It provides:

- Data ingestion from databases, files, APIs, and applications.
- Data routing and transformation.
- Monitoring of data flows.

- Error handling and retry mechanisms.
- Back-pressure to control data flow.
- Data provenance to track where data came from.

Architecture:

Source → NiFi → Validate/Transform → HDFS/Database/Cloud

NiFi is particularly useful when an organization has **many different data sources and destinations**.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:

The major failure points include **DataNodes, NameNodes, NodeManagers, disks, and network connections**.

Handling failures:

1. **DataNode failure:**
HDFS automatically creates new replicas from healthy copies.
2. **NameNode failure:**
Use **NameNode High Availability** with Active and Standby NameNodes.
3. **Disk failure:**
Use multiple disks and HDFS replication.
4. **NodeManager failure:**
YARN detects the failed node and reschedules tasks on healthy nodes.
5. **Network failure:**
Use redundant network connections and proper rack-aware architecture.

Overall approach:

Failure Detection → Recovery → Replication/Rescheduling → Monitoring

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:

I would design the pipeline with security at every stage.

Architecture:

Data Sources → Secure NiFi → Validation → HDFS → Processing → Analytics

Security measures:

- Use **TLS/SSL** for data in transit.

- Use **Kerberos** for authentication in Hadoop.
- Apply role-based access control.
- Encrypt sensitive data at rest.
- Validate and sanitize incoming data.
- Maintain audit logs and data provenance.
- Use network segmentation and firewalls.
- Implement backup and disaster recovery.

Example:

A database sends data through an encrypted connection to NiFi. NiFi validates and routes it to HDFS, where only authorized users can access the data.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:

I would explain it using a simple example:

"Imagine we have a huge warehouse containing millions of boxes. Instead of asking one person to handle every box, we divide the boxes among many workers. Each worker processes a portion of the boxes at the same time. If one worker is unavailable, another can handle the work."

In Hadoop:

- **HDFS** is the warehouse where data is stored across many machines.
- **YARN** is the manager that distributes computing resources.
- **MapReduce/Spark** are the workers that process the data.
- **Replication** keeps backup copies of important data.
- Adding more machines allows the system to handle more data.